Alzheimer's Disease Classification from Brain MRI Scans Using a Custom Convolutional Neural Network: Design, Implementation, and Deployment in a Mobile Health Application

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Abstract: Alzheimer's disease is a progressive neurodegenerative disorder characterized by cognitive decline, memory loss, and behavioral impairment. Early diagnosis remains a critical challenge due to the subtle onset of symptoms and the reliance on expert-driven analysis of medical imaging. In recent years, artificial intelligence (AI), particularly deep learning [1], has shown significant potential in improving diagnostic accuracy and enabling early-stage detection.This paper presents an AI-driven framework for the early prediction of Alzheimer's disease using brain Magnetic Resonance Imaging (MRI). The core of the proposed system is a Convolutional Neural Network (CNN) model trained on a large dataset of labeled MRI scans. The model is designed to automatically extract discriminative features from brain images and classify them into four stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. To enhance model robustness and generalization, preprocessing techniques such as normalization, resizing, and data augmentation were applied. The proposed model achieves high classification performance, demonstrating strong capability in identifying early-stage Alzheimer's patterns that are often difficult to detect through traditional methods. In addition, the system integrates an intelligent AI-based chatbot utilizing natural languageprocessing techniques to provide users with real-time medical guidance and disease-related information, further extending the role of AI beyond diagnosis to patient support.The overall framework highlights the effectiveness of combining deep learning-based medical image analysis with AI-powered interaction systems. This approach not only improves early detection accuracy but also contributes to accessible, scalable, and user-centered healthcare solutions. The results indicate that AI can play a transformative role in assisting clinicians and enhancing decision-making in neurodegenerative disease diagnosis.

Keywords—Alzheimer's, MRI images, CNN, Medical Image Classification, Early Detection, MRI Preprocessing, Data Augmentation, TensorFlow, AI Chatbot

INTRODUCTION

Alzheimer's disease is a progressive neurodegenerative disorder [2] that leads to a continuous decline in cognitive functions, including memory, reasoning, and decision-making. Despite significant advancements in medical imaging technologies, early diagnosis of Alzheimer's disease remains a complex and challenging task. Traditional diagnostic approaches rely heavily on manual interpretation of brain imaging data, particularly Magnetic Resonance Imaging (MRI), which requires extensive expertise, is time-consuming, and may be prone to human error

In recent years, artificial intelligence (AI) has emerged as a transformative solution in the field of medical diagnostics. Among various AI techniques, deep learning—especially Convolutional Neural Networks (CNNs)[3], [4] — has demonstrated exceptional capability in analyzing medical images and extracting complex patterns that are often imperceptible to the human eye [5], [6], [7]. CNN-based models can automatically learn hierarchical feature representations from MRI scans, enabling accurate classification and early detection of neurodegenerative diseases [5],[8] such as Alzheimer.

This research proposes an AI-driven system for the early prediction of Alzheimer's disease based on MRI image analysis. The core contribution of this work lies in the design and implementation of a deep learning model that processes brain MRI scans and classifies them into four distinct stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Unlike conventional methods, the proposed approach eliminates the need for manual feature extraction by leveraging automated feature learning, thereby improving both efficiency and diagnostic accuracy.

To further enhance the performance of the AI model, several preprocessing and optimization techniques are applied, including image normalization, resizing, and data augmentation. These techniques help improve the generalization capability of the model and reduce the risk of overfitting. The system is trained and evaluated on a structured dataset of MRI images, achieving high classification accuracy and demonstrating strong potential for real-world medical applications.

In addition to image-based diagnosis, the proposed system incorporates an AI-powered conversational module based on natural language processing (NLP). This component enables users to interact with the system, obtain medical insights, and receive guidance regarding Alzheimer's disease. By integrating deep learning for image analysis with AI-driven user interaction, the system extends beyond diagnosis to provide a comprehensive intelligent healthcare solution. Deep learning techniques have gained considerable attention because of their ability to learn complex representations directly from raw medical images without requiring handcrafted feature extraction. This capability is particularly valuable in brain MRI analysis, where subtle structural changes may be difficult to identify through conventional image processing techniques.

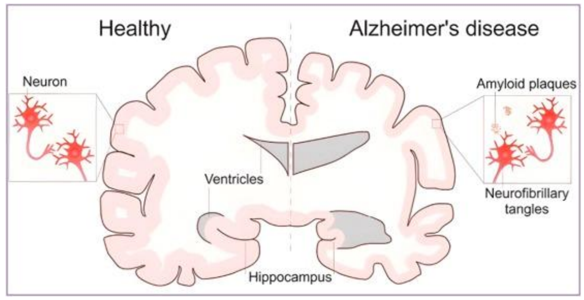


Figure 1

Overall, this work highlights the critical role of artificial intelligence in advancing early detection of Alzheimer's disease. The proposed AI-based framework not only improves diagnostic precision but also contributes to scalable, accessible, and user-centric healthcare systems, paving the way for future innovations in medical AI applications.

LITERATURE SURVEY

Over the years, several researchers have used different approaches to diagnose this disea. The following paragraphs give a quick insight into the works completed to date.

Suresha et al. [5] used a rectified adam optimizer and a deep neural network to classify images into Normal, AD, and MCI, respectively. They achieved a high accuracy of 99.5% by using a Histogram of Oriented Gradients to extract features.

Lan [9] defined a new model that would help doctors detect Alzheimer's. He made use of diffusion tensor images which are used to deploy brain networks. Those details were captured by him using the graph theory method. He then applied three different algorithms to check the accuracy and suggested a better approach. The algorithms used by him were SVM, random forest, and CNN trained with SPL features.

The accuracy of the three models is 90%, 98%, and 90%, respectively. The sensitivity of the model was 92%,96% and 72% while specificity was 88%,100%,94% respectively

Zubair [10] claimed a method to detect Alzheimer's disease. He made use of a 5 stage ML pipeline process for the detection in which each stage had a sub-stage. Multiple classifiers were applied to this pipeline. He concluded that the random forest Classifier had better performance metrics.

Khan et al. [10] made use of the Random-Forest classifier to compare the performance in imputation and non-imputation methods. They observed that the imputation method gives 87% accuracy, and the non-imputation method gives 83% accuracy. It further classified the subjects as demented or non-demented, respectively.

Asim et al. [6] proposed a multi-atlas approach to detect Alzheimer's disease by utilizing the unique features extricated from every atlas template and the joined characteristics of the two atlases using PCA and casted-off SVM for classification. They achieved 94% accuracy for AD vs. CN, 76.5% accuracy for CN vs. MCI, and 75.5% accuracy for MCI vs. AD. They observed that the multi-atlas approach showed better results than the single-atlas approach.

Alam [11] stated in his paper that early-stage detection can prevent the spread of disease. He made use of structural magnetic resonance(MRI) for capturing brain images from the database. He postulated the use of kernels for projecting the data onto the available linear space. He then applied a Support Vector Machine(SVM) for classifying the data. He obtained a good accuracy of 93.85% for his classification with high sensitivity and specificity.

Moein [7] first applied voxel morphometry analysis for capturing some of the most crucial MRI images. He then carried out a principal component analysis on the features extracted. Then a hybrid manifest learning framework was presented in the given subspace. Then, a label propagation model was applied to classify into two types as normal and mild by taking a chunk of training data. The model provided a whooping accuracy of 93.86% for the given classification.

Kumar Lama [8] made a collaborative approach to distinguish AD from different other diseases. He made use of SMR's to determine AD amongst mild cognitive impairment and Health Control. He used three algorithms, namely RELM,SVM and IVM, for this segregation process. In addition to that, a discriminative approach including kernels is provided to tackle various data distributions which are complex. He concluded from his classification approach that RELM had better performance metrics.

Dr. Bryan [12] saw that varieties among intersite and multi vendor estimations restricted AI applications for cerebral bloodstream imaging strategies in Alzheimer's Disease. Such types can be vigorously standardized in human vision but need significant advances in figuring out how to dodge perils from overlooked and underrepresented measurable blunders.

Escudero et al. [13] proposed a ML approach using biomarkers in their paper. They tested a personalized classifier for the disease using a method learning locally weighted and biomarkers. The methodology attempts to classify the subject first and then later decides which biomarker to order. They classified the patients who were MCI who advanced to AD inside a year against the individuals who didn't.

Proposing Of Our Work

Deep Learning is known to be learning a hierarchical set of representations such that it learns low mid and high-level features. Deep neural networks can adapt to more complex data sets. It's better in generalizing previously unseen data because of its multiple layers. Different algorithms use Deep Learning's fundamental expertise and use diverse datasets to train and test these algorithms.

This application is an artificial intelligence-driven system for the early detection and classification of the disease using brain (MRI). The core of the proposed work is a deep learning-based framework that utilizes a (CNN) to automatically analyze MRI scans and identify patterns associated with different stages of the disease. The system is designed to provide accurate, efficient, and scalable diagnostic support, reducing reliance on manual interpretation.

The overall workflow of the proposed system consists of several key stages: data acquisition, preprocessing, model training, classification, and result interpretation. Initially, a dataset of labeled MRI brain images is collected, representing four categories of Alzheimer’s progression: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. These categories enable fine-grained classification and support early-stage detection. like neurons in humans, deep learning has layers that help the model or algorithm learn and

process the data. These layers process the data given to them as the input and learn by processing the input while traveling through the layers. When it passes out the last layer, an activation function is applied at last, and finally, we get the predicted output from the deep learning model.

Dataset

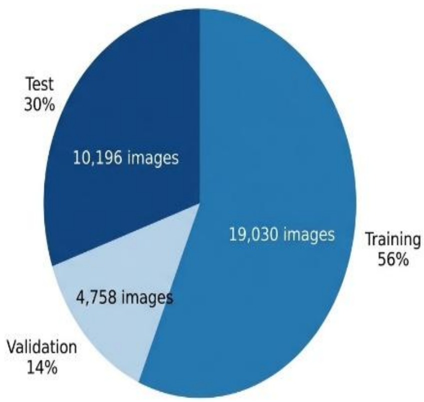


Figure 2

The dataset used in this research is obtained from an open-source medical imaging repository available on Kaggle [14]. It consists of approximately 40,000 brain Magnetic Resonance Imaging

(MRI) images, categorized into four distinct classes representing different stages of the disease: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

The dataset is structured to support supervised deep learning, where each image is labeled according to its corresponding disease stage. To ensure reliable training and evaluation of the

proposed model, the dataset is divided into two main subsets using an 70:30 ratio. Specifically, 70% of the data is allocated for training, while the remaining 30% is reserved for testing and validation purposes.

This split enables the model to learn meaningful patterns during the training phase and evaluate its performance on unseen data during the testing phase. Maintaining a consistent distribution of classes across both training and testing sets is essential to avoid bias and ensure fair evaluation. Therefore, the dataset is carefully partitioned to preserve class balance, ensuring that each category is proportionally represented in both subsets. The use of a publicly available dataset also enhances the reproducibility of the research and allows fair comparison with previously published studies.

Furthermore, prior to model training, the dataset undergoes a preprocessing stage to improve data quality and model performance. This includes resizing images to a fixed input dimension, normalizing pixel values, and applying data augmentation techniques such as

rotation, flipping, and zooming. These steps increase dataset variability and enhance the model's ability to generalize to new, unseen MRI scans.

The use of a publicly available dataset ensures reproducibility and allows for consistent

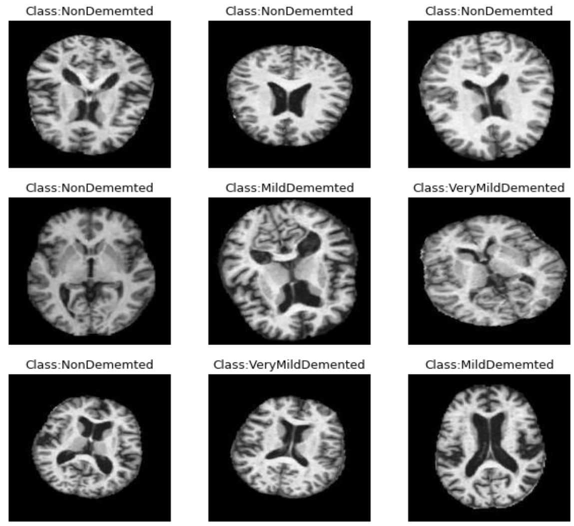


Figure 3

benchmarking with existing studies. Additionally, employing the same dataset distribution across training and testing phases guarantees that the evaluation of the proposed model is conducted under controlled and unbiased conditions.

Methodology

The proposed methodology follows a structured artificial intelligence pipeline designed for accurate classification of the disea using MRI images. The process begins with data acquisition from the prepared dataset, followed by preprocessing and feature extraction using a deep learning model.

Initially, the training dataset is fed into the model where the model learns hierarchical feature representations directly from images. The learning process involves extracting low-level features such as edges and textures, followed by high-level structural patterns associated with brain degeneration.

After the training phase, the model is evaluated using the testing dataset to measure its generalization capability. Performance metrics such as accuracy are calculated to assess the effectiveness of the model in correctly classifying unseen data. The classification output is divided into four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

The system architecture integrates multiple components, including the dataset module,

preprocessing unit, the model, and evaluation module. These components work together in a unified pipeline, where data flows sequentially from input acquisition to final prediction. The trained model is then deployed to classify new images in real time, providing fast and reliable diagnostic results.

Overall, the methodology emphasizes the use of deep learning for automated feature extraction and classification, ensuring high accuracy and efficiency in the disea detection.

CNN

The core component of the proposed system is a (CNN) model specifically designed for the classification of the disea using brain images. It is one of the convolutional neural system models, consisting of 9 convolutional layers. The model is developed to automatically extract hierarchical features from medical images and accurately classify them into four stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

The model operates through a hierarchical feature learning process:

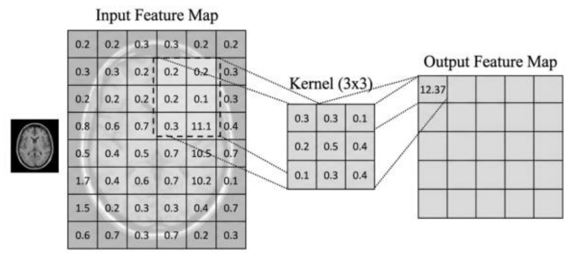
1- Feature Extraction Stage: The convolutional layers scan the brain images using filters to detect important patterns such as edges, textures, and structural variations in brain regions.
2- Feature Reduction Stage: Pooling layers reduce the dimensionality of the extracted features while preserving the most relevant information.
Feature Learning Stage: Deeper layers combine low-level features into more complex representations, allowing the model to distinguish between different stages of the disease.
3- Classification Stage: The fully connected layers interpret the learned features and map them into class probabilities using the Softmax function.

model to distinguish between different stages of the disea.

4.ClassifcationStage

The fully connected layers interpret the learned features and map them into class probabilities using the Softmax function.

Model Architecture:



$$\text{Output}(i,j) = \sum_m \sum_n \text{Input}(i+m, j+n) \times \text{Kernel}(m,n)$$

Figure 4

Convolution operation: a 3×3 kernel slides across the input feature map,
computing element-wise multiplication and summation to produce the output feature map."

Model layers:

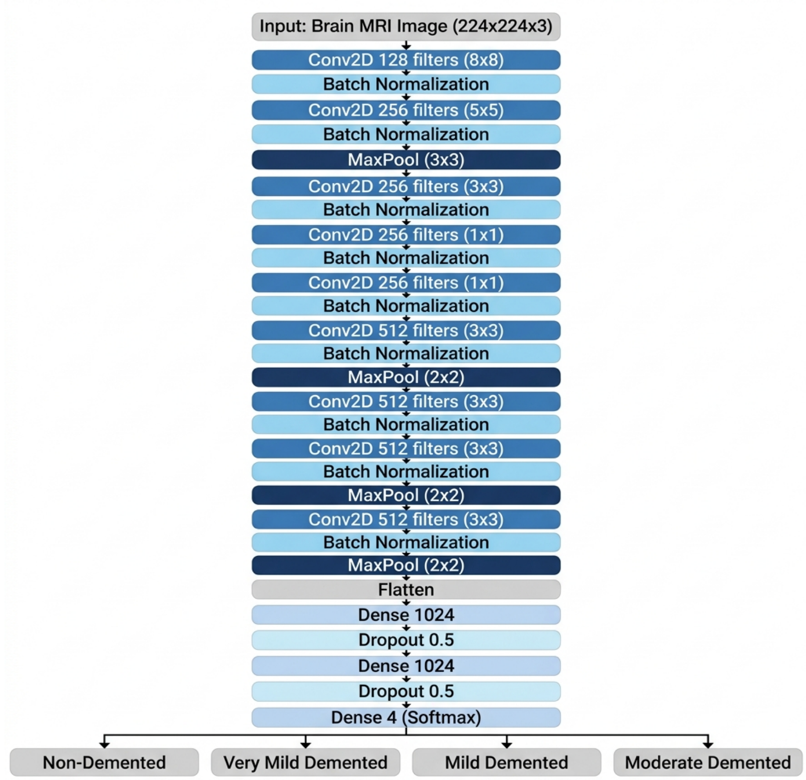


Figure 5

First Convolution Layer: Applies 128 filters of size 8×8. Stride (3×3) reduces spatial dimensions fast. Extracts high-level features from the input MRI image. ReLU activation introduces non-linearity.Batch Normalization [16]: Normalizes feature maps, stabilizes training, and speeds up convergence.

Second Convolution Layer: Applies 256 filters of size 5×5. "Same" padding keeps output size fixed. Learns more detailed features.

Batch Normalization [16]: Same role: improves training stability.

Max Pooling: Reduces spatial size, keeps important features only, and reduces computation [3].

Third Convolution Layer: Applies 256 filters of size 3×3.

1×1 Convolution Layers [3]: Used for feature refinement, reduces dimensional complexity, and helps combine features without changing image size

Fourth Convolution Layer: Increases depth (512 filters) and learns more complex patterns. Max Pooling: Reduces feature map size and prevents overfitting [17].

Deep Convolution Layers: Capture high-level abstract features, important for classification accuracy. Max Pooling: Further reduces dimensions [3].

Final Convolution Block: Final feature extraction stage. Max Pooling: Reduces data before flattening.

Flatten Layer: Converts 2D feature maps → 1D vector, preparing data for Dense layers.

First Dense Layer (1024 neurons): Fully connected layer that learns high-level patterns. Dropout [17]: Randomly disables 50% of neurons to prevent overfitting.

Second Dense Layer (1024 neurons): Adds more learning capacity. Dropout: Same purpose: reduce overfitting [17].

Output Layer: • 4 neurons → 4 classes • Softmax converts outputs → probabilities[1]

IMPLEMENTATION

The model was implemented using Python and deep learning frameworks such as TensorFlow and Keras [18] .

The model architecture was designed as a sequential network consisting of multiple Convolutional, pooling, and fully connected layers. Each layer was carefully configured to optimize feature extraction from brain images and improve classification performance.

The dataset was first loaded and preprocessed using standard image processing techniques. Images were resized to a fixed input size and normalized to ensure consistency across all samples. Data augmentation techniques, including rotation, flipping, and zooming, were applied during training to enhance model generalization and reduce overfitting.

The model was trained on the prepared dataset using supervised learning. During training, categorical cross-entropy was used as the loss function [1], while the Adam optimizer was employed for efficient gradient-

based optimization. The training process was performed over multiple epochs, with performance monitored using validation accuracy and loss metric.

The dataset was split into training and testing sets using an 70:30 ratio. The training set was used to fit the model, while the testing set was used to evaluate its performance on unseen data. Key evaluation metrics included accuracy, precision, recall, and F1-score.

To further improve performance, techniques such as dropout and early stopping were applied. Dropout layers were used to prevent overfitting by randomly deactivating neurons during training, while early stopping was employed to halt training once validation performance stopped improving.

The trained model demonstrated strong classification capability across all four disea stages, indicating its effectiveness in detecting both early and advanced patterns in MRI images.

RESULTS DISCUSSION

The model has performance evaluation and provides a detailed discussion of the obtained results. The evaluation focuses on the model’s ability to accurately classify brain images into four stages of the disease.

From a clinical perspective, the ability of the model to accurately identify early stages of Alzheimer's disease is particularly important. Early diagnosis enables timely medical intervention, continuous monitoring, and improved patient care planning. The high classification performance achieved by the proposed system suggests that artificial intelligence.

Author & Year	Data	Technology / Model	Accuracy
Smith et al., 2019	MRI (ADNI dataset)	RNN(Recurrent Neural Network)	C-index ≈ 0.78
Desikan et al., 2022	OASIS clinical dataset	Decision Tree – SVM – Random Forest	80% - 86.9%
Saoud & AlMarzouqi, 2024	MRI (ROI-based brain regions)	3DVisionTransformer +DeepBelief Network	88% - 92%
Multi-modalStudies 2024	MRI + PET	CNN + Transfer Learning	87%
Our Project	MRI	CNN	93%

Figure 6

Performance Metrics

To assess the effectiveness of the proposed model, several standard evaluation metrics were used, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of the model's classification performance, particularly in multi-class medical image analysis.The model achieved high overall accuracy on the test dataset, demonstrating its strong capability in distinguishing between different stages of the disea. Precision and recall values across most classes were consistently high, indicating that the model is effective in both correctly identifying positive cases and minimizing false negatives. The F1-score further confirms the balance between precision and recall, highlighting the robustness of the model.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Figure 7

Training Performance Analysis

During the training process, the model exhibited stable convergence behavior. The training and validation accuracy curves showed a steady increase over epochs, while the loss curves demonstrated a consistent decrease. This indicates that the model successfully learned meaningful patterns from the brain image dataset.

Additionally, the gap between training and validation performance remained minimal, suggesting that overfitting was effectively controlled. This can be attributed to the use of data augmentation, dropout layers, and early stopping techniques, which enhanced the generalization capability of the model.

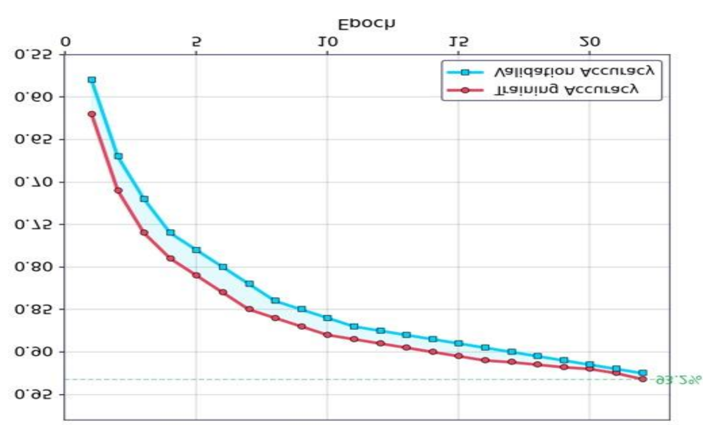


Figure 8

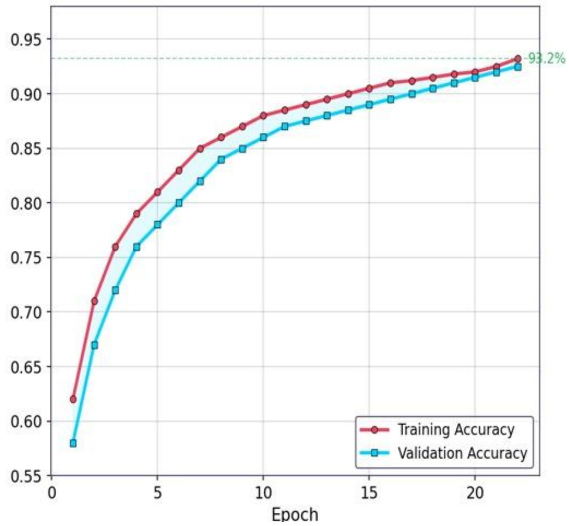


Figure 9

Classification Analysis

The model demonstrated strong performance across all four classes:

- **Non-Demented:** High accuracy due to distinct structural characteristics in MRI scans.
- **Very Mild Demented:** Slightly more challenging to classify, as early-stage features are subtle and less pronounced.
- **Mild Demented:** Achieved reliable classification with clear differentiation from normal cases.
- **Moderate Demented:** High classification accuracy due to significant structural changes in brain regions.

The results indicate that the model is particularly effective in identifying advanced stages of Alzheimer's disease, while maintaining reasonable accuracy in detecting early-stage conditions.

Model Effectiveness

The strong performance of the proposed model can be attributed to its ability to automatically extract

hierarchical features from brain images. Unlike traditional machine learning approaches, the model does not rely on manual feature engineering, allowing it to capture complex spatial patterns associated with neurodegeneration.

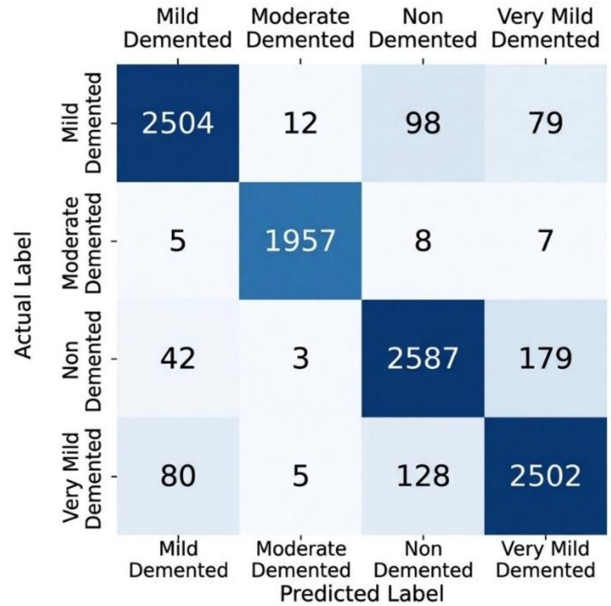


Figure 10

Furthermore, the use of preprocessing techniques and balanced dataset distribution contributed significantly to improving model accuracy and stability. The integration of deep learning enabled the system to achieve a high level of diagnostic reliability.

Application Flow

The application flow begins with user interaction and extends to AI-based diagnosis, cognitive support, and continuous progress monitoring. As illustrated in *Figure 1*, the system follows a structured pipeline consisting of authentication, data processing, AI inference, and user feedback modules.

Overall System Workflow

- **User Authentication:** The process starts when the user logs into the mobile application using secure credentials. This ensures safe access to personal health data and system features.

- **MRI Image Upload:** The user uploads a brain MRI image through the application interface. The image is transmitted securely to the backend server via API requests.

- **Data Preprocessing Module:** As shown in *Figure 2*, the uploaded MRI image undergoes preprocessing steps such as resizing, normalization, and formatting to match the CNN model input requirements.

- **AI Model Inference (CNN):**

The preprocessed image is passed to the trained as illustrated in *Figure 3*. The model extracts hierarchical features and performs classification into four categories:

- Non-Demented
- Very Mild Demented
- Mild Demented
- Moderate Demented

- **Result Generation and Visualization**

The prediction result, along with probability scores, is returned to the backend and displayed to the user in a clear and interpretable format. The system may also provide basic guidance based on the predicted stage.

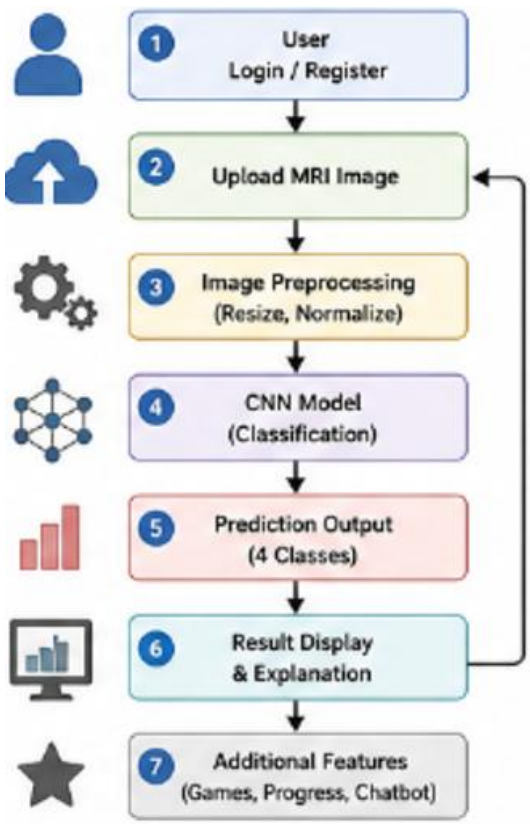


Figure 9

Cognitive Support and Monitoring Flow

Beyond diagnosis, the system incorporates AI-supported modules to assist users in managing their condition and improving cognitive function.

Brain Games Module:

As illustrated in *Figure 4*, users can access a set of memory-enhancing brain games designed to stimulate cognitive activity. These games are developed to improve memory, attention, and problem-solving skills. The system may track user interactions and performance during gameplay.

The module includes a variety of interactive games that target different cognitive functions such as memory recall, concentration, pattern recognition, logical reasoning, and reaction speed. Each game is developed with varying levels of difficulty to accommodate users with different cognitive abilities and provide a personalized experience.

Progress Tracking Module

The application includes a progress monitoring feature, as shown in *Figure 5*, where users can track their cognitive performance over time. Data from AI predictions and game performance are stored and analyzed to provide insights into user improvement and behavioral patterns. The module stores historical records of MRI classification results, brain game performance, and user interaction data within a secure database. By maintaining a comprehensive history, the system can generate meaningful insights regarding cognitive improvement, stability, or decline. These insights support informed decision-making and help users better understand their cognitive health status.

Daily Routine and Activity Support

The system may also include structured routines to help users maintain consistency in daily activities. This contributes to better cognitive stability and overall patient management. The module provides personalized reminders for essential daily activities, including medication schedules, physical exercise, hydration, sleep routines, and cognitive training sessions. These reminders help users maintain consistency in their daily lives and reduce the likelihood of forgetting important tasks, which is a common challenge among individuals affected by cognitive decline.

AI Chatbot Interaction
AI Chatbot Module:

- As shown in Figure 6, users can interact with an AI-powered chatbot The chatbot provides: system performance.
- General health guidance and support
- Information about the disea

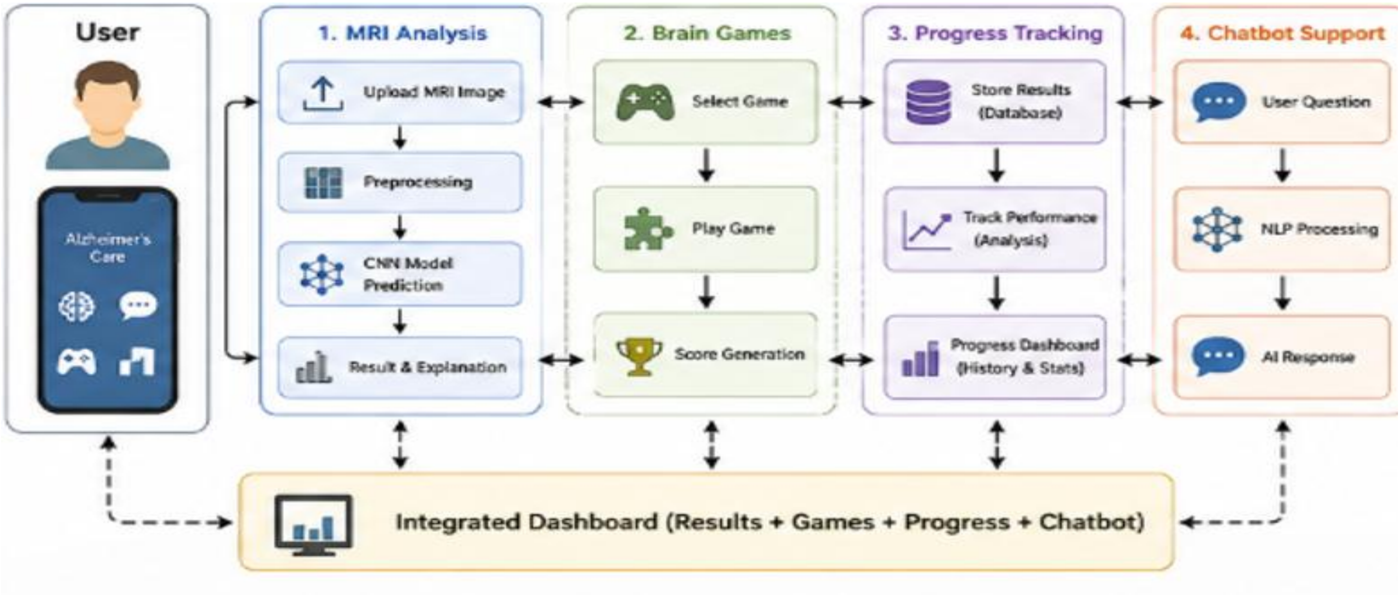


Figure 10

System Architecture Flow

The system architecture consists of the following interconnected components:

- **Mobile Application (Frontend):** Handles user interaction, image upload, game interface, and progress visualization
- **Backend Server:** Manages API communication, preprocessing, and data flow
- **CNN Model (AI Engine):** Performs feature extraction and classification
- **Database:** Stores user data, MRI results, game scores, and progress history
- **Chatbot Module:** Provides AI-based conversational interaction

The overall workflow begins at the mobile application layer, where users interact with the system through an intuitive user interface. User requests, MRI image uploads, and activity data are securely transmitted to the backend server for processing. The backend acts as the central coordinator, managing data flow, authentication, preprocessing operations, and communication with the AI components..

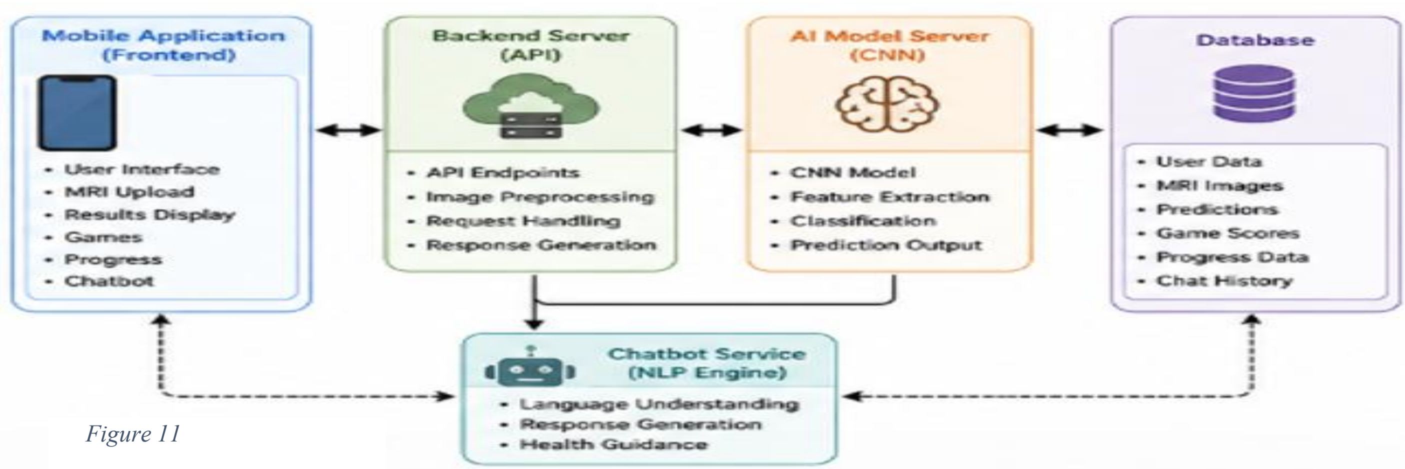


Figure 11

CONCLUSION AND FUTURE WORKS

future work will focus on enhancing the AI model and expanding the system capabilities. This includes:

- Incorporating larger and more diverse datasets to improve model generalization
- Exploring advanced deep learning architectures such as transfer learning and transformer-based models
- Integrating multimodal data sources (e.g., MRI, PET, and clinical data) for improved diagnostic accuracy
- Improving explainability through Explainable AI (XAI) techniques to increase trust in model predictions
- interaction and personalization
- Expanding the progress tracking system using predictive analytics to forecast cognitive decline

In conclusion, the proposed system demonstrates the significant potential of artificial intelligence in transforming Alzheimer's disease diagnosis and management. By combining deep learning, user

interaction, and continuous monitoring, this work contributes to the development of intelligent, scalable, and accessible healthcare solutions.

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